Exam SF2940 21.10.2022

1a) $X \mid A=a \in U(0,a)$, hence $f_{X|A=a}(x) = \begin{cases} \frac{1}{a} & \text{if } 0<x<a, \\ 0 & \text{else.} \end{cases}$

Thus $$\mathbb{E}[X|A=a] = \int_{\mathbb{R}} \frac{x}{a} \mathbb{1}_{\{0<x<a\}} dx$$
$$= \frac{1}{a}\int_0^a x\,dx = \frac{1}{2a} x^2 \Big|_0^a = \frac{a}{2}.$$

Therefore $\mathbb{E}[X|A] = \frac{A}{2}$.

Next, by the law of total expectation

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X|A]\big] \overset{\text{above}}{=} \frac{1}{2}\mathbb{E}[A].$$

Now by the given distribution of the r.v. A, we have

$$\mathbb{E}[A] = \int_{\mathbb{R}} a \cdot 6a(1-a)\mathbb{1}_{\{0<a<1\}} da$$
$$= \int_0^1 6a^2 da - \int_0^1 6a^3 da$$
$$= 2a^3\Big|_0^1 - \frac{6}{4}a^4\Big|_0^1 = 2 - \frac{3}{2} = 1/2.$$

Answer: $\mathbb{E}[X|A] = \frac{A}{2}$, $\mathbb{E}[X] = \frac{1}{2}\mathbb{E}[A] = 1/4$. //

16.) By definition of conditional probability density

$$f_{X|A=a}(x) = \frac{f_{X,A}(x,a)}{f_A(a)}, \quad 0<a<1.$$

Rewriting this we find

$$f_{X,A}(x,a) = f_{X|A=a}(x)\, f_A(a)$$

To obtain $f_X(x)$, the marginal, we integrate out A:

$$f_X(x) = \int_{\mathbb{R}} f_{X,A}(x,a)\,da = \int_{\mathbb{R}} f_{X|A=a}(x) f_A(a)\,da$$

mixing integral

$$= \int_{0<x<1} \frac{1}{a}\,\mathbb{1}_{\{0<x<a\}} \cdot 6a(1-a)\,\mathbb{1}_{\{0<a<1\}}\,da$$

$$= 6\int_x^1 (1-a)\,da = 6\,\frac{1}{2}(1-a)^2\Big|_1^x = 3(1-x)^2.$$

$\mathbb{1}_{\{x<a<1\}}$

Answer: $f_X(x) = 3(x-1)^2\,\mathbb{1}_{\{0<x<1\}} = \begin{cases} 3x^2-6x+3, & \text{if } 0<x<1 \\ 0, & \text{else.} \end{cases}$

2a.) From the formulary, the characteristic function of $N \in Po(\lambda)$ is given by $\varphi_N(t) := \mathbb{E}[e^{itN}] = e^{\lambda(e^{it}-1)}$.

Thus $\varphi_{\frac{N}{\lambda}}(t) = \mathbb{E}[e^{i\frac{t}{\lambda}N}] = e^{\lambda(e^{\frac{it}{\lambda}}-1)}$.

As $\lambda \to \infty$, we have the Taylor expansion

$$\lambda\left(e^{\frac{it}{\lambda}}-1\right) = \lambda\left(1 + \frac{it}{\lambda} + O\left(\frac{1}{\lambda^2}\right) - 1\right)$$
$$= it + O\left(\frac{1}{\lambda}\right).$$

Thus $\lim_{\lambda\to\infty} \varphi_{\frac{N}{\lambda}}(t) = e^{it}$, which is the characteristic function of $\delta(1)$. But convergence in distribution to a constant implies convergence in probability. Thus $\frac{N}{\lambda} \xrightarrow{P} 1$, as $\lambda \to \infty$ //

2b.) The characteristic function of $\frac{N-\lambda}{\sqrt{\lambda}}$ is given by

$$\varphi_{\frac{N-\lambda}{\sqrt{\lambda}}}(t) = \mathbb{E}\left[e^{it\left(\frac{N-\lambda}{\sqrt{\lambda}}\right)}\right] = e^{-it\sqrt{\lambda}} \cdot \mathbb{E}\left[e^{\frac{it}{\sqrt{\lambda}}N}\right]$$
$$\overset{2a)}{=} e^{-it\sqrt{\lambda}}\, e^{\lambda(e^{\frac{it}{\sqrt{\lambda}}}-1)}.$$

2b.) con't: A Taylor expansion now yields

$$\lambda\left(e^{\frac{it}{\sqrt{\lambda}}}-1\right) = \lambda\left(1+\frac{it}{\sqrt{\lambda}}-\frac{t^2}{2\lambda}+\frac{o(1)}{\lambda}-1\right)$$

as $\lambda\to\infty$. $\qquad = \sqrt{\lambda}it - \frac{t^2}{2} + o(1)$.

goes to zero as λ goes to infinity for any fixed t.

Hence $\varphi_{\frac{N-\lambda}{\sqrt{\lambda}}}(t) = e^{-it\sqrt{\lambda}+\sqrt{\lambda}it-\frac{t^2}{2}+o(1)}$

$$= e^{-t^2/2+o(1)},$$

as $\lambda\to\infty$. Thus by continuity of the exponential function,

$$\lim_{\lambda\to\infty}\varphi_{\frac{N-\lambda}{\sqrt{\lambda}}}(t) = e^{-t^2/2}.$$

By the uniqueness theorem, $\frac{N-\lambda}{\sqrt{\lambda}} \xrightarrow[\lambda\to\infty]{d} N(0,1)$. //

2c.) The characteristic function of $\frac{S_N}{\sqrt{\lambda}}$ is given by

$$\varphi_{\frac{S_N}{\sqrt{\lambda}}}(t) = \mathbb{E}\left[e^{it\frac{S_N}{\sqrt{\lambda}}}\right] = \varphi_{S_N}\left(\tfrac{t}{\sqrt{\lambda}}\right)$$

composition formula, X_i's i.i.d.

$$= g_N\left(\varphi_{X_1}\left(\tfrac{t}{\sqrt{\lambda}}\right)\right), \quad g_N \text{ probability generating function of } N.$$

$$= e^{\lambda\left(\varphi_{X_1}\left(\frac{t}{\sqrt{\lambda}}\right)-1\right)}.$$

2c.) cont. The characteristic function $\varphi_{X_1}\left(\frac{t}{\sqrt{\lambda}}\right)$ has the Taylor expansion

$$\varphi_{X_1}\left(\frac{t}{\sqrt{\lambda}}\right) = 1 + \overbrace{\mathbb{E}X_1}^{=0}\frac{it}{\sqrt{\lambda}} + \frac{1}{2}\overbrace{\mathbb{E}X_1^2}^{=2}\frac{(it)^2}{\lambda} + o\left(\frac{1}{\lambda}\right)$$

$$= 1 - \frac{t^2 + o(1)}{\lambda},$$

as $\lambda \to \infty$.

Thus, $$\lambda\left(\varphi_{X_1}\left(\frac{t}{\sqrt{\lambda}}\right) - 1\right) = \lambda\left(1 - \frac{t^2 + o(1)}{\lambda} - 1\right)$$

$$= -t^2 + o(1),$$

as $\lambda \to \infty$, and moreover

$$\varphi_{\frac{S_N}{\sqrt{\lambda}}}(t) = e^{\lambda\left(\varphi_{X_1}\left(\frac{t}{\sqrt{\lambda}}\right) - 1\right)} = e^{-t^2 + o(1)},$$

as $\lambda \to \infty$. By continuity of the exponential function

$$\lim_{\lambda \to \infty} \varphi_{\frac{S_N}{\sqrt{\lambda}}}(t) = e^{-t^2} = e^{-\frac{2 \cdot t^2}{2}},$$

which is the characteristic function of a $N(0, 2)$ r.v. By the uniqueness theorem, $\frac{S_N}{\sqrt{\lambda}}$ converges in distribution to an $N(0,2)$ random variable.

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3 a) W a Wiener process.

$$\mathbb{E}[W(1)^2 + W(2) \mid W(1)] = \underbrace{\mathbb{E}[W(1)^2 \mid W(1)]}_{W(1)^2} + \underbrace{\mathbb{E}[W(2) \mid W(1)]}_{\text{Since } W(t) \text{ is B.M.}} \underbrace{= W(1)}$$

$$= W(1)^2 + W(1). \quad //$$

3 b.) Since W is a Gaussian process, $(W(1), W(2), W(3) - W(1))'$ is a Gaussian vector. The mean vector is zero because

$\mathbb{E}[W(1)] = \mathbb{E}[W(2)] = \mathbb{E}[W(3)] = 0$.

Next, we note that $\mathbb{E}[W(1)^2] = 1$, $\mathbb{E}[W(2)^2] = 2$ and $\mathbb{E}[(W(3) - W(1))^2] = 2$ since $W(t)$ is a BM.

Moreover, $\mathbb{E}[W(2)W(1)] = 1$ and $\mathbb{E}[(W(3) - W(1))(W(1))] = 0$, since $W(3) - W(1)$ is independent from $W(1)$.

Finally, $\mathbb{E}[(W(3) - W(1))W(2)] = \underbrace{\mathbb{E}[W(3)W(2)]}_{=2} - \underbrace{\mathbb{E}[W(2)W(1)]}_{=1}$

$= 1$.

Thus the covariance matrix is $\Lambda = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$.

Alternatively, $\begin{pmatrix} W(1) \\ W(2) \\ W(3) - W(1) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}}_{=: B} \begin{pmatrix} W(1) \\ W(2) \\ W(3) \end{pmatrix}$.

Then $\Lambda = B \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix} B'$. //

Answer: $N(0, \Lambda)$.

3c.) By definition we have,

$$\mathbb{E}[X(t)|W(s)] = \mathbb{E}\left[e^{W(t)-\frac{t}{2}}|W(s)\right]$$

$$= e^{-\frac{t}{2}}\mathbb{E}\left[e^{W(t)-W(s)+W(s)}|W(s)\right]$$

$$= e^{-\frac{t}{2}}\mathbb{E}\left[e^{W(t)-W(s)}e^{W(s)}|W(s)\right]$$

$$= e^{-\frac{t}{2}}e^{W(s)}\mathbb{E}\left[e^{W(t)-W(s)}|W(s)\right].$$

Since $W(t)-W(s)$ is independent of $W(s)$, we have

$$\mathbb{E}\left[e^{W(t)-W(s)}|W(s)\right] = \mathbb{E}\left[e^{W(t)-W(s)}\right].$$

Next, $W(t)-W(s) \in \mathcal{N}(0, t-s)$, thus with the hint (or explicit computation)

moment generating function $\Rightarrow = \psi_{W(t)-W(s)}(1) \quad (r=1)$

$$= e^{\frac{(t-s)}{2}}.$$

Hence, $\mathbb{E}[X(t)|W(s)] = e^{W(s)}e^{-\frac{t}{2}}e^{\frac{t-s}{2}}$

$$= e^{W(s)-\frac{s}{2}}.$$

$$= X(s).$$

Answer: $\mathbb{E}[X(t)|W(s)] = X(s)$, $t \geq s \geq 0$.

4.) First, by the (strong) law of large numbers,

$$\frac{1}{n}\sum_{i=1}^{n}(X_i)^2 \xrightarrow[n\to\infty]{a.s.} \mathbb{E}[X_1^2] \text{ since } \mathbb{E}[X_i^2] \le \mathrm{Var}(X_i) = \sigma^2 < \infty.$$

Now, $\mathbb{E}[X_i^2] = \mathrm{Var}(X_i) + \mathbb{E}[X_i]^2 = \sigma^2 + \mu^2$.

Second, $\frac{1}{n}S_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow[n\to\infty]{a.s.} \mu$, since $\mathbb{E}|X_i| < \infty$, by the LLN.

As a.s. convergence implies convergence in probability and distribution, we have by Cramér-Slutsky that

$$\frac{1}{n^2}(S_n)^2 = \left(\frac{1}{n}S_n\right)\left(\frac{1}{n}S_n\right) \xrightarrow[n\to\infty]{d} \mu^2,$$

and, thus, $\frac{1}{n}\sum_{i=1}^{n}(X_i)^2 - \frac{1}{n^2}(S_n)^2 \xrightarrow[n\to\infty]{d} \sigma^2 + \mu^2 - \mu^2 = \sigma^2$.

Since convergence in distribution to a constant implies convergence in probability, we get

$$\frac{1}{n^2}\left[n\sum_{i=1}^{n}(X_i)^2 - (S_n)^2\right] \xrightarrow[\text{as } n\to\infty]{p} \sigma^2.$$

This takes care of the denominator.

The numerator is now,

$$n^{1/2}\left(\exp\left(\frac{S_n}{n}\right) - \exp(\mu)\right) = n^{1/2}\left(g\left(\frac{S_n}{n}\right) - g(\mu)\right)$$

where $g(x) := \exp(x)$.

4 cont.) In the exercise 9, we showed that

$$\sqrt{n}\left(g\left(\frac{S_n}{n}\right) - g(\mu)\right) \xrightarrow[n\to\infty]{d} N\left(0, (g'(\mu)\sigma)^2\right),$$

Since $\frac{S_n}{n} \xrightarrow[n\to\infty]{P} \mu$ and $g = \exp$ is smooth. Moreover, $g'(\mu) = e^{\mu}$.
Alternatively, a Taylor expansion in combination with Cramér-Slutsky will yield the same conclusion.
This takes care of the numerator.

Combining the above, we have

$$n^{1/2}\,\frac{\exp\left(\frac{S_n}{n}\right) - \exp(\mu)}{n\sum_{i=1}^{n}(X_i)^2 - (S_n)^2} = \frac{n^{1/2}\left(\exp\left(\frac{S_n}{n}\right) - \exp(\mu)\right)}{\frac{1}{n^2}\left(n\sum_{i=1}^{n}(X_i)^2 - (S_n)^2\right)}$$

$$\xrightarrow[n\to\infty]{d} N\left(0, e^{2\mu}\sigma^2\right).$$

$$\xrightarrow[n\to\infty]{P} \sigma^2.$$

$$\xrightarrow[n\to\infty]{d} \frac{N\left(0, e^{2\mu}\sigma^2\right)}{\sigma^2} = N\left(0, \frac{e^{2\mu}}{\sigma^2}\right),$$

where we used once again Cramér-Slutsky.

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5a) $\mathbb{E}[X(t)\,|\,X(s)] = \mathbb{E}[\underbrace{X(t)-X(s)}\,|\,X(s)] + \underbrace{\mathbb{E}[X(s)\,|\,X(s)]}$

increment $X(t)-X(s)$ is indep from $X(s)=X(s)-X(0)$. Property of cond' expectation Properties 1.) & 2.)

$= X(s)$

$= \mathbb{E}[X(t)-X(s)] + X(s)$.

$X(t)-X(s) \sim P_o(t-s)$, hence $\mathbb{E}[X(t)-X(s)] = t-s$

$= t-s+X(s)$. //

5b.) $R_X(t,s) \overset{\text{def}}{=} \mathbb{E}[X(t)X(s)]$.

Assume first that $t>s$, then

$\mathbb{E}[X(t)X(s)] \overset{\text{l.t.e.}}{=} \mathbb{E}[\mathbb{E}[X(t)X(s)\,|\,X(s)]]$

property of cond' expectation

$= \mathbb{E}[X(s)\,\mathbb{E}[X(t)\,|\,X(s)]]$

5a)

$= \mathbb{E}[X(s)^2] + \mathbb{E}[X(s)](t-s)$

Since $X(s) \sim P_o(s)$, we have $\mathbb{E}[X(s)]=s$, $Var(X(s))=s$

$\Rightarrow \mathbb{E}[X(s)^2] = Var(X(s)) + \mathbb{E}[X(s)]^2 = s+s^2$.

$= s+s^2+s(t-s)$

$= s+st$, for $t>s$.

Next, assume that $t \le s$, then similarly to the above

$\mathbb{E}[X(t)X(s)] = \mathbb{E}[X(t)\,\mathbb{E}[X(s)\,|\,X(t)]]$

$= \mathbb{E}[X(t)^2] + \mathbb{E}[X(t)](s-t)$

$= t+t^2+t(s-t)$

$= t+ts$, for $t \le s$.

[b) cont] Combining both cases, we get

$$R_{\underline{X}}(t,s) = E[X(t)X(s)] = \begin{cases} s+st, & t>s \\ t+st, & t\leq s \end{cases}$$

$$= st + \min(s,t).$$

Answer: $R_{\underline{X}}(t,s) = st + \min(s,t)$, $t,s \geq 0$. //

[c) For a r.v. Y taking values in the non-negative integers, the probability generating function is defined as

$$g_Y(r) := E[r^Y] \quad \text{(and exists for at least } |r|<1\text{)}.$$

Thus, we have

$$g_{X(t)}(r) = E[r^{X(t)}]$$

$X(t) \in P_o(t)$

$$= \sum_{n=0}^{\infty} r^n e^{-t} \frac{t^n}{n!}$$

$$= e^{-t} \sum_{n=0}^{\infty} \frac{(rt)^n}{n!} = e^{-t} e^{rt} = e^{t(r-1)},$$

which is well-defined for all $r \in \mathbb{R}$.

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